Problem Set 1 - IC104

1 Questions

Essential Mathematics for Economic Analysis, 4th edition
Page 9. Question 5: Expand and simplify:

(a) $\frac{p^{24}p^3}{p^4p}$ (b) $\frac{a^4b^{-3}}{(a^2b^{-3})^2}$ (c) $\frac{3^4(3^2)^6}{(-3)^{15}3^7}$ (d) $\frac{p^\gamma(pq)^\sigma}{p^{2\gamma+\sigma}q^{\sigma-2}}$

Page 14. Question 6: Expand each of the following:

(a)$(x+2y)^2$ (b) $(\frac{1}{x}-x)^2$ (c) $(3u-5v)^2$ (d) $(2x-5w)(2x+5w)$

Page 18. Question 2 a) b) c) d) e) f): Simplify the various expressions:

(a) $\frac{x}{10}-\frac{3x}{10}+\frac{17x}{10}$ (b) $\frac{9a}{10}-\frac{a}{2}+\frac{a}{5}$ (c) $\frac{b+2}{10}-\frac{3b}{15}+\frac{b}{10}$

(d) $\frac{x+2}{3}+\frac{1-3x}{4}$ (e) $\frac{3}{2b}-\frac{5}{3b}$ (f) $\frac{3a-2}{3a}-\frac{2b-1}{2b}+\frac{4b+3a}{6ab}$

Page 22. Question 5 a) b) c): Solve for x:

(a) $\sqrt{x}=9$

(b) $\sqrt{x}\cdot\sqrt{4}=4$

(c) $\sqrt{(x+2)}=25$

Page 45. Question 1 a) c) e):

Solve the following quadratic equations, if they have solutions:

(a) $15x-x^2=0$ (c) $(q-3)(q+4)=0$
(e) $x(x+1)=2x(x-1)$

Page 47. Question 2: Solve the following simultaneous equations:

(a) $5x + 2y = 3$; $2x + 3y = -1$

(b) $x - 3y = -25$; $4x + 5y = 19$

(c) $2x + 3y = 3$; $6x + 6y = -1$

Page 54. Question 3 a) b) e):

Express these sums in summation notation:
(a) $4 + 8 + 12 + 16 + \cdots + 4n$

(b) $1^3 + 2^3 + 3^3 + 4^3 + \cdots + n^3$

(e) $3x + 9x^2 + 27x^3 + 81x^4 + 243x^5$

Page 98. Question 2:

Find the equilibrium price for each of the two linear models of supply and demand:

(a) $D = 75 - 3P$; $S = 20 + 2P$

(b) $D = 100 - 0.5P$; $S = 10 + 0.5P$